

PRST Class 10/09/2020

Classical definition: all the outcomes are equally probable.

Empirical probability -

subjective probability

Ω : collection of all possible outcomes of an experiment

Ω - sample space.

Event - a subset of Ω .

P : We define probability as a real-valued function if A is any event of Ω

(i) $P(A) \geq 0$

(ii) $P(\Omega) = 1$

(iii) $A_1, A_2, \dots$ are disjoint events then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

$(A_i \cap A_j = \emptyset \Rightarrow A_i \text{ and } A_j \text{ are disjoint})$
mutually exclusive.

Result 1 $P(\emptyset) = 0$

proof:

$$A_i = \emptyset \quad i=1, 2, \dots$$

$$A_i \cap A_j = \emptyset$$

$$\bigcup_{i=1}^{\infty} A_i = \emptyset$$

$$- \quad P(\emptyset) = P(\emptyset) + P(\emptyset) + \dots$$

$$\Rightarrow P(\emptyset) = 0$$

Result 2:

$A_1, A_2, \dots, A_n$ are disjoint events.

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i)$$

proof:

$$A_{n+1} = \emptyset$$

$$A_{n+2} = \emptyset \dots$$

$$A_1 \cup A_2$$

$$A_i = \emptyset \quad i = n+1, n+2, \dots$$

$$\bigcup_{i=1}^{\infty} A_i = \bigcup_{i=1}^n A_i \cup \emptyset = \bigcup_{i=1}^n A_i$$

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i) = \sum_{i=1}^n P(A_i)$$

Result 3.

$$P(A^c) = 1 - P(A)$$

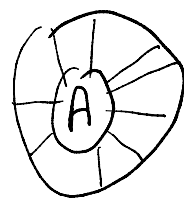
$$A \cup A^c = A^c$$

$$- \quad P(A \cup A^c)$$

Lemma.

$$P(\Omega) = 1 = P(A_1 \cup A_2) = P(A \cup A^c).$$

$$\Rightarrow P(A) = 1 - P(A^c)$$



Result 4.

$$A \subseteq B$$

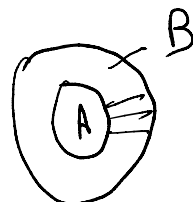
$$P(A) \leq P(B).$$

proof

$$B = A \cup (B-A)$$

$$P(B) = P(A) + P(B-A) \geq 0$$

$$P(A) \leq P(B).$$



Result 5 :

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Result 6 : $P\left(\bigcup_{i=1}^m A_i\right) = \sum_{i=1}^m P(A_i) - \sum_{i_1 < i_2} P(A_{i_1} \cap A_{i_2})$

$$+ \sum_{i_1 < i_2 < i_3} P(A_{i_1} \cap A_{i_2} \cap A_{i_3})$$

$$+ (-1)^{m-1} P(A_1 \cap A_2 \cap \dots \cap A_m).$$